Raft: Understandable Distributed Consensus

Node States: Follower, Candidate, Leader.

Heartbeat intervals: Leaders send AppendEntries RPCs every 50ms.

Election timeout: Randomized between 150ms and 300ms to avoid split votes.

Majority quorum: A cluster of 2F+1 nodes tolerates F Byzantine/crash failures.

Log Replication & Safety Invariants

Leader Append-Only property: Leaders never overwrite or truncate their own logs.

Log Matching property: Matching term and index implies identical preceding log entries.

Commit rule: An entry is committed once replicated across majority nodes by current leader.